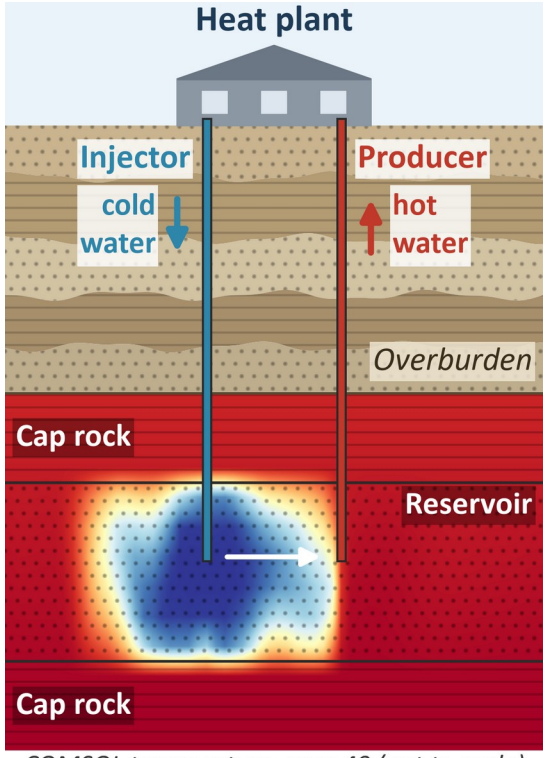
Neural operator surrogate for 3D geothermal doublet simulation

1 Geothermal doublet (COMSOL)



COMSOL temperature, year 40 (not to scale)

25 50 75 100
Temperature (°C)

2 Dataset and surrogate

Dataset	COMSOL
Scenarios	1,535 (LHS design)
Split	1,074 / 229 / 232
Inputs	8 maps + 14 params
Outputs	3D T, p; 3 well curves
Horizon	40 years

Parameters → FiLM

3D U-Net

4D Fourier neural operator

Well-output head

Model A
data loss

Model B
+ mass, energy

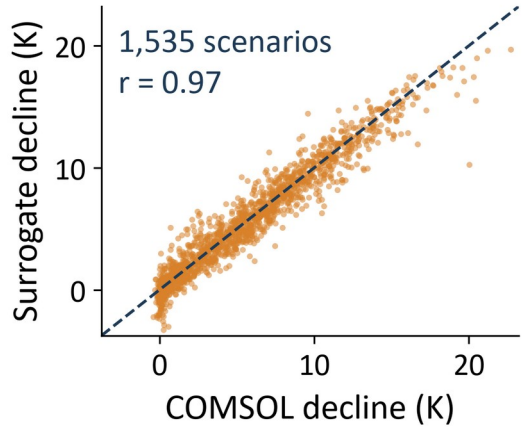
3 Accuracy on 232 test scenarios

Model A	MAE
Temperature field	0.92 K
Pressure field	0.019 MPa
Cooled cells (> 1 K)	3.0 K
Production temp.	0.95 K
40-year cooling	1.26 K
Thermal power	0.29 MW

Rel. error	A	B
Temperature	0.24%	0.24%
Pressure	0.08%	0.19%
Wells	4.90%	4.74%

Physics loss: no gain.
COMSOL's own fields do not satisfy its residuals.

4 Screening of 1,535 scenarios



40-year production-temperature decline

Capacity, yr 40	MWth
Cluster A / C / B	24.8 / 22.6 / 20.0
All: surrogate	22.3
All: COMSOL	22.4

No scenario exceeds 10% decline (median 5.0 K).

A compact neural operator trained on COMSOL data reproduces 3D fields, well outputs and ensemble statistics at ≈ 0.05 s per scenario, supporting preliminary geothermal screening; a physics loss should be validated against the simulator before use.